

COMPSCI 701:
Creating Maintainable Software
Lecture 1.1: Introduction

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Agenda

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- Course Overview
- Course Details
- Workload
- Time Management
- Learning Journals
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- Use of AI
- Assignment 1
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- Quality
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- Key Points

- Admin
 - Course Rep nominations
- Course Overview
- Assignment 1
- Maintainable Software — what is it and why do we care?
- Lecture preparation
 - Program Comprehension Models recording (Canvas)
 - Context and Design Schema recording (Canvas)
 - One persons view of what makes code unmaintainable
(<https://www.mindprod.com/jgloss/unmain.html>)

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- in-depth look as to what it means for software to be maintainable
- principles and practices that have been proposed for developing maintainable software
- examine their efficiency and effectiveness — not just understand what they are but evaluate and critique them
- look at relevant research

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People

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More details on Canvas

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Course Practice

- There will be code! (Java)
- Expect to write—and read—code
- There will be in-class team exercises.
- Typically one day a week will be mostly such exercises, but other days will also have shorter exercises
- Participation in the exercises is considered a crucial component for succeeding in this course

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Communication

- Meetings — lectures
- Resources
 - Canvas — lab handouts, assignment handouts, probably other stuff
- Course marks — Canvas
- Email — Electronic Mail is an official and the primary means of communication with students
 - **Please use email to communicate with us — not Ed Discussion or Canvas**
- Journal
 - **Bound** book(s) (stapled sheets is ok)
 - Handwritten notes or printed copies of electronic notes
 - Must contain only content created by you. **No copies of lecture slides, web pages, pages of books, or anything you did not create**
 - May be taken into tests
 - Can be used to record: lecture notes, thoughts, questions, work log
- Ed Discussion
 - general questions and discussion
 - Do not post answers to assignment questions
 - Posts should show respect to all who might read them

Workload

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- An average student is expected to spend, on average, 10 hours a week in order to get an average grade
 - to get a better than average grade an average student may need to spend more than 10 hours a week
 - scheduled lectures make up 3 of those hours $\Rightarrow$ if you do not attend lectures you should expect to make up those hours in other ways

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Time Management

- You are responsible for meeting the stated deadlines
- The deadlines are posted in several places. You are expected to manage your time so that you can meet them.
- “I have something else due on that day so can I have an extension” indicates poor time management skills and so shows a lack of professionalism

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Learning Journals

- written notes and commentary to support learning
- research shows that taking notes recording what you have seen or heard (e.g. lectures) aids learning
- research also shows that writing commentary reflecting on what you have seen or heard aids learning
- many different ideas on what a learning journal looks like, search and find something that works for you

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Course Assessment

- 100% coursework (no exam)
- 3 assignments (10% each)
- 1 assignment (the last, 20%)
- Participation (10% Lectures, Discussions, Research studies)
- 2 Tests in weeks 6 and 12 (20% each)

Use of AI

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- Use of AI for invigilated assessment (i.e. Tests) is prohibited
- Use of AI in any form for other assessments is **allowed**
 - you must take responsibility for everything in your submission
 - the tests will be to assess what you have **learned**
 - “Here is the assignment description. Create a complete solution”
⇒ learning == 0
 - “Here is my implementation. Change to make it more maintainable”
⇒ learning essentially 0 (how can you be sure what is produced is consistent with what is discussed in class)
 - “Here is my implementation. List possible maintainability issues it might have.”
⇒ Possibly much learning **provided each issues suggested by the AI is checked against course material**

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Assignment 1: Maintainable Kalah

- Implement a simple program appropriate as an assignment for an introductory course on Object-Oriented Programming course in Java (the board game Kalah)
- Your implementation should be as “maintainable” as possible
- Provide evidence as to how maintainable your implementation is

Kalah

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- Played with 48 seeds on a board with 2 stores and 12 houses
- Each player controls 1 store and 6 houses
- A game starts with 4 seeds in each house
- Game play
 - A turn consists of 1 or more moves
 - A move consists of taking seeds from a house and planting 1 in each house clockwise, including the player's store (but not opponent's)
 - If a move ends in the player's store, the player gets another move
 - If a move ends in player's house that was empty, the player gets that seed and all seeds in the opposing house
 - The game ends when one player only has empty houses
 - The score is all the seeds in a player's store and all houses (highest score wins)
- *Kalah is marketed in some places as "Mancala"*

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Kalah Example

```

+---+---+---+---+---+---+---+---+
|  0 | 1[ 4] | 2[ 4] | 3[ 4] | 4[ 4] | 5[ 4] | 6[ 4] | P2 |
|    |-----+-----+-----+-----+-----+-----+
| P1 | 6[ 4] | 5[ 4] | 4[ 4] | 3[ 4] | 2[ 4] | 1[ 4] |  0 |
+---+---+---+---+---+---+---+---+

```

Player P1's turn - Specify house number or 'q' to quit: 3

```

+---+---+---+---+---+---+---+---+
|  1 | 1[ 4] | 2[ 4] | 3[ 4] | 4[ 4] | 5[ 4] | 6[ 4] | P2 |
|    |-----+-----+-----+-----+-----+
| P1 | 6[ 5] | 5[ 5] | 4[ 5] | 3[ 0] | 2[ 4] | 1[ 4] |  0 |
+---+---+---+---+---+---+---+---+

```

Player P1's turn - Specify house number or 'q' to quit: 5

```

+---+---+---+---+---+---+---+---+
|  2 | 1[ 5] | 2[ 5] | 3[ 5] | 4[ 4] | 5[ 4] | 6[ 4] | P2 |
|    |-----+-----+-----+-----+-----+
| P1 | 6[ 6] | 5[ 0] | 4[ 5] | 3[ 0] | 2[ 4] | 1[ 4] |  0 |
+---+---+---+---+---+---+---+---+

```

Player P2's turn - Specify house number or 'q' to quit: 6

```

+---+---+---+---+---+---+---+---+
|  2 | 1[ 5] | 2[ 5] | 3[ 5] | 4[ 4] | 5[ 4] | 6[ 0] | P2 |
|    |-----+-----+-----+-----+-----+
| P1 | 6[ 6] | 5[ 0] | 4[ 5] | 3[ 1] | 2[ 5] | 1[ 5] |  1 |
+---+---+---+---+---+---+---+---+

```

Player P1's turn - Specify house number or 'q' to quit:

Kalah Example

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Player P1's turn - Specify house number or 'q' to quit: 4

...

Player P2's turn - Specify house number or 'q' to quit: 5

```
+-----+-----+-----+-----+-----+-----+-----+-----+
|  3  | 1[ 6] | 2[ 6] | 3[ 5] | 4[ 4] | 5[ 0] | 6[ 1] | P2 |
|      |-----+-----+-----+-----+-----+-----+-----|
| P1 | 6[ 7] | 5[ 1] | 4[ 0] | 3[ 1] | 2[ 6] | 1[ 6] | 2 |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

Player P1's turn - Specify house number or 'q' to quit: 3

```
+-----+-----+-----+-----+-----+-----+-----+-----+
|  9  | 1[ 6] | 2[ 6] | 3[ 0] | 4[ 4] | 5[ 0] | 6[ 1] | P2 |
|      |-----+-----+-----+-----+-----+-----+-----|
| P1 | 6[ 7] | 5[ 1] | 4[ 0] | 3[ 0] | 2[ 6] | 1[ 6] | 2 |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

Player P2's turn - Specify house number or 'q' to quit:

...

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Test Infrastructure

```
public class Kalah {
    public static void main(String[] args) {
        new Kalah().play(new MockIO());
    }
    public void play(IO io) {
        io.println("+-----+-----+-----+-----+-----+-----+-----+-----+");
        io.println("| 0 | 1[ 4] | 2[ 4] | 3[ 4] | 4[ 4] | 5[ 4] | 6[ 4] | P2 |");
        io.println("|      |-----+-----+-----+-----+-----+-----+-----+");
        io.println("| P1 | 6[ 4] | 5[ 4] | 4[ 4] | 3[ 4] | 2[ 4] | 1[ 4] | 0 |");
        io.println("+-----+-----+-----+-----+-----+-----+-----+-----+");
        io.println("Player P1's turn - Specify house number or 'q' to quit:");
    }
}
```

- MockIO provides input from keyboard and output to console

Test Infrastructure

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```
...
io.setExpected("test/simple_start.txt");
io.println("+----+-----+-----+-----+-----+...");
...
```

- MockIO can compare I/O actually done with expected I/O specified by a file
- E.g., in test/simple_start.txt:

```
# Do one move, which doesn't affect the player's store, then quit.
>+----+-----+-----+-----+-----+-----+-----+-----+
>|  0 | 1[ 4] | 2[ 4] | 3[ 4] | 4[ 4] | 5[ 4] | 6[ 4] | P2 |
>|    |-----+-----+-----+-----+-----+-----+-----|
>| P1 | 6[ 4] | 5[ 4] | 4[ 4] | 3[ 4] | 2[ 4] | 1[ 4] |  0 |
>+----+-----+-----+-----+-----+-----+-----+-----+
>Player 1's turn - Specify house number or 'q' to quit:
<1
...
```

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Restrictions

- No Java 8 (or later) features, including JDK (see explanation and resources on Canvas)
- No reference to any other files (e.g. use of configuration files).
- No use of third-party libraries.
- Do not use `System.exit()` under any circumstances.

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- COMPSCI701 is using GitHub classrooms
- You will need to create a GitHub account
- Clone the supplied GitHub repository for the assignment to get the starter kit for the assignment
- More details available in the Canvas assignment description, and documentation in the starter kit
- **Commit early and often to your repo. Do **not** wait until you have something “working” before doing your first commit!**
- **Provide useful commit messages to remind you of what you did (in particular, any use of AI)**
- **Push to `submission` branch to trigger Github Continuous Integration Workflow that will runs all tests**

Assessment

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- Passes all tests (COMPSCI701 is a graduate course. The assignment is suitable for a *first* assignment in object-oriented development at *second* year.)
- Object-oriented design (“good” **classes** in addition to `kalah.Kalah`, at least 3 of which should have the “right” **number of objects** constructed from them during the execution of your implementation)
- Code quality must meet minimum levels
- Maintainable—does your code have characteristics known to create problems for maintainability
- **Bonuses** — early submission, demonstration of experimentation
- **Penalties** — late submission, wrong submission branch, development on submission branch
- **Details on Canvas**

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Deliverables

- Make sure your GitHub repository submission branch is up-to-date before the due date.
- Keep the same source file structure as in the starter pack (i.e. all source in src)
- Due **Friday 8 August 2359hrs**

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Maintainability: Why should we care?

- The *lifetime cost* of a product is the total of all expenses (whether measured in money, effort, time, or some other way) over the life of the product (from when it is first conceived until it is retired)
- *Software maintenance* refers to the activities that take place to keep a software product useful
- Studies show that the majority of the lifetime expenses occur after the first delivery of the product
- $\Rightarrow$ reducing the cost of maintenance activities will significantly lower the lifetime cost of software products
- the *maintainability* of a software product is the ease with which maintenance activities can be carried out on it
- “maintainability” is one aspect of the software’s *quality*

Maintainability versus Maintenance

- Maintenance — the activities that take place after delivery of the product.
Typically described as:
 - corrective — remove defects
 - adaptive — change the product to a changed environment
 - preventative — improving some aspect of its quality
 - perfective — improving some aspect of its usefulness, including adding new features

many of the activities are carried out before the delivery!
- Maintainability — the property or attribute of the product that affects the ease with which maintenance activities can be carried out

this property is strongly influenced by how the code is designed and written

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Why do developers care about quality?

- “quality” for developers typically means “be more efficient at creating software”
- e.g.
 - produce code cheaper
 - produce code faster
 - produce code with less effort
 - produce code with fewer defects
 - deploy products faster
 - respond to change requests faster
 - fix faults faster
 - fix faults with less effort

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What might “Quality” mean?

- Maintainability
- Comprehensibility
- Alterability
- Testability
- Extensibility
- Understandability
- Readability
- Modifiability
- Portability
- Buildability
- Changeability
- Modularity
- Traceability
- Security
- Scalability
- Performance
- Availability
- Reliability
- ...

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Quality Attributes

- Sometimes called “non functional requirements”, except there can be non functional requirements that are not quality attributes.
- quality — the standard of something as measured against other things of a similar kind; the degree of excellence of something (Oxford Dictionary)
- some notion of how “good” the product is $\Rightarrow$ “satisfies the needs of its stakeholders”
- A **quality attribute** is a [quantifiable] or testable property of a system that is used to indicate how well the system satisfies the needs of its stakeholders

Bass, Clements, and Kazman (2013) “Software Architecture in Practice” (3rd ed) Addison-Wesley

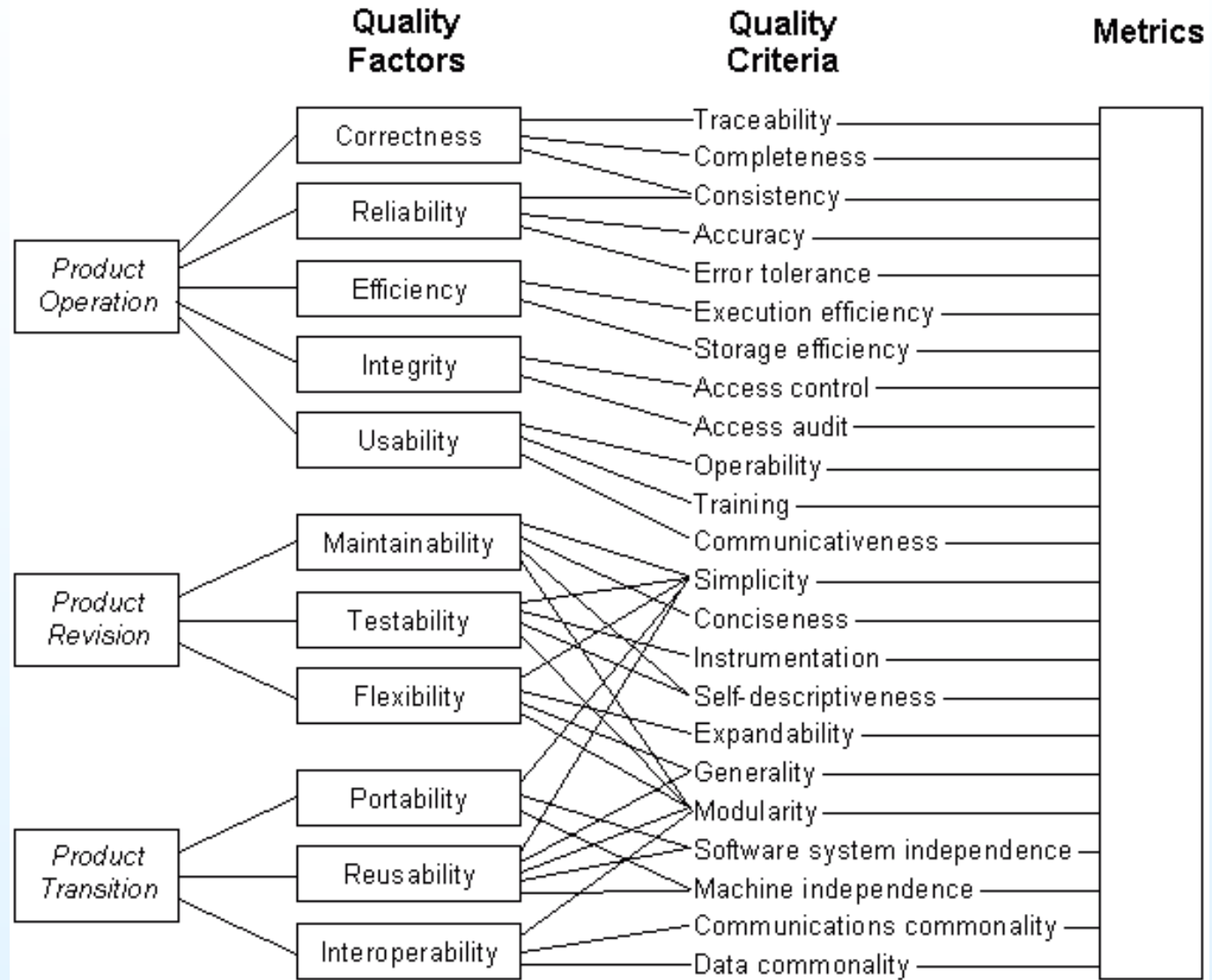
Defining Quality

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- There have been various attempts to define “quality” in general, and quality attributes in particular, over the years
 - McCall’s software quality model (1977)
 - Boehm’s software quality model (1978)
 - FURPS software Quality Model (1987)
 - ISO/IEC 9126 (1991,2001)
 - ISO/IEC 25010 (2011)

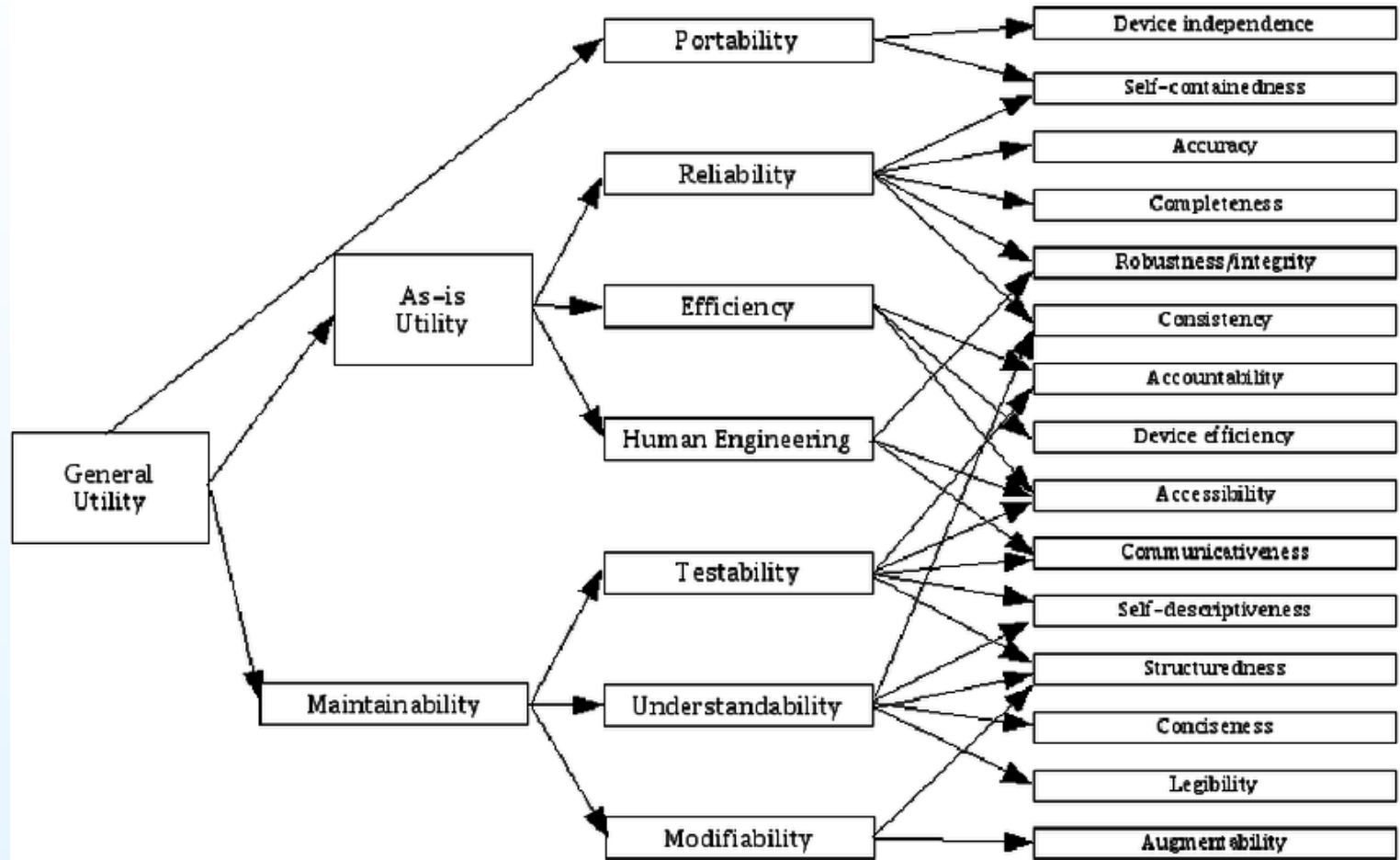
McCall's Software Quality Model

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Boehm's Software Quality Model

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Defining Quality Attributes: Maintainability

McCall the effort required to locate and **fix a fault** in the program within its operating environment.*

IEEE Standard Glossary of Software Engineering Terminology, 1990

the ease with which software can be **maintained, enhanced, adapted, or corrected** to satisfy specified requirements*

ISO/IEC 9126/25010 The degree of effectiveness and efficiency with which a product or system can be **modified to improve it, correct it or adapt it** to changes in environment, and in requirements

Boehm's Quality Model **Maintainability** is composed of Testability, Understandability, and **Modifiability***

Random Web Source **Modifiability** encompasses two aspects: **Maintainability** and Flexibility*

(* quotes not necessarily dead accurate)

ISO/IEC 25010

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Maintainability The degree of effectiveness and efficiency with which a product or system can be modified to improve it, correct it or adapt it to changes in environment, and in requirements It is composed of the following sub-characteristics:

- Modularity
- Reusability
- Analysability
- Modifiability
- Testability

Course Definition for Maintainability

- The degree of **effectiveness and efficiency** with which **maintenance activities** can be carried out on the product. It includes these **independent** sub-attributes:
 - **Comprehensibility (Analysability)**†—Degree of effectiveness and efficiency with which the implementation can be understood in order to conduct maintenance activities with confidence
 - **Alterability (Modifiability)**†—degree to which a product or system can be effectively and efficiently changed without introducing defects or degrading existing product quality.
 - **Testability**— degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and tests can be performed to determine whether those criteria have been met.
- **Notes**
 - As much as possible, arguments about each sub-attribute should not refer to any of the others. E.g., an argument about alterability must assume everything necessary about the code is understood
 - There is no mention of “after delivery” in these definitions — “maintenance activities” can occur before delivery

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Key Points

- Read the Syllabus on Canvas
- Maintenance costs dominate the lifetime cost of a software product
- The more maintainable the software, the lower the maintenance costs, and so the lower the lifetime cost
- Maintainability (for this course) consists of Comprehensibility, Alterability, and Testability